

Bengali LLM Quality Evaluation Study

Bangladesh Locale — Portfolio Project Overview

MD. IMRAN HOSSAIN (IMH) | Independent Human Evaluator

Project Overview

An independent, self-directed human evaluation of 50 Bengali LLM outputs designed to assess linguistic, contextual, instructional, and cultural quality within a Bangladesh linguistic and cultural context. The study uses a structured QA framework to identify substantive and non-material issues.

Evaluation Snapshot

50 Cases	10 Categories	44 Pass (88%)	6 Fail (12%)
----------	---------------	---------------	--------------

Methodology

- Model: Gemini Web — “3.6 Flash”; evaluation date: 1 October 2026; language: Bengali; locale: Bangladesh.
- Five controlled prompts were evaluated in each of 10 categories: Grammar, Spelling, Bangladesh Localization, Bangladesh vs West Bengal, Translation Fidelity, Meaning Preservation, Factuality, Instruction Following, Tone/Register, and Formatting/Terminology.
- Outputs were assessed using a documented error taxonomy and severity framework.
- The framework distinguishes source-text errors from model errors, accepts legitimate Bengali variation, and does not penalize relevant extra information when the prompt does not prohibit it.

Key Findings

- 44 of 50 cases (88%) were classified as Pass; 6 (12%) as Fail.
- Sampled Grammar, Spelling, Bangladesh Localization, Factuality, Instruction Following, and Meaning Preservation cases all passed.
- Material failures were concentrated in cultural comparison, translation naturalness, meaning/context alignment, tone/register, and terminology consistency.
- The study demonstrates that Bengali LLM QA requires more than surface grammar checking; local context, semantic precision, register, and instruction fidelity also require human review.

Representative Cases

Case	Issue identified
BQA-018	Bangladesh–West Bengal cultural overgeneralization
BQA-022	Unnatural primary translation despite a correct alternative
BQA-041	Formal Bengali fluency and redundancy
BQA-043	Contextual/semantic mismatch in customer-support wording
BQA-045	Tone shift from request to mandatory instruction
BQA-047	Terminology consistency issue

Deliverables

50-case evaluation dataset (Excel) • Final Evaluation Report • Representative case analysis • Reproducibility record

Independent portfolio project | Bengali LLM quality evaluation | Bangladesh locale | 1 October 2026